B.Sc. (Hons.) in Environmental Science

B.Sc. (Hons.) in Environmental Science is an undergraduate degree program that focuses on the study of the environment and its interactions with living and non-living components. The program typically lasts for three years and covers a wide range of topics, including:

1. Ecology and biodiversity
2. Environmental chemistry and toxicology
3. Climate change and meteorology
4. Natural resource management
5. Environmental policy and law
6. Environmental impact assessment
7. Sustainable development and green technology

The program aims to provide students with a strong foundation in the natural sciences, as well as an understanding of the social, economic, and political factors that influence environmental issues. Graduates of the program may pursue careers in environmental consulting, conservation, research, education, or policy-making.

Unit 1 - Environment

Definition

The environment is the sum total of all the physical, chemical, and biological factors that surround and influence an organism. It includes both biotic and abiotic components.

Types of Environment

There are two main types of environment: natural and man-made. Natural environments include forests, oceans, mountains, and other natural landscapes. Man-made environments include cities, towns, and other human-made structures.

Components of Environment

The environment is made up of both biotic and abiotic components. Biotic components include living organisms such as plants, animals, and microorganisms. Abiotic components include non-living factors such as air, water, and soil.

Segments of Environment

The environment can be divided into four segments: atmosphere, hydrosphere, lithosphere, and biosphere. The atmosphere is the layer of gases that surrounds the Earth. The hydrosphere includes all the water on Earth. The lithosphere is the solid outer layer of the Earth. The biosphere includes all living organisms on Earth.

Scope and Importance

The environment is important because it provides the conditions necessary for life to exist. It also provides resources that are essential for human survival and development. The scope of environmental studies includes the study of the natural world and the impact of human activities on the environment.

Need for Public Awareness

Public awareness is important for the protection and conservation of the environment. It is necessary for people to understand the importance of the environment and the impact of their actions on it. Public awareness can be increased through education and outreach programs.

Ecosystem

Definition

An ecosystem is a community of living organisms interacting with each other and their physical environment.

Types of Ecosystem

There are many different types of ecosystems, including terrestrial, aquatic, and man-made ecosystems. Terrestrial ecosystems include forests, grasslands, and deserts. Aquatic ecosystems include oceans, lakes, and rivers. Man-made ecosystems include cities and agricultural lands.

Structure of Ecosystem

An ecosystem is made up of both biotic and abiotic components. The biotic components include the living organisms in the ecosystem, while the abiotic components include the physical and chemical factors that influence the ecosystem.

Food Chain

A food chain is a sequence of organisms in which each organism feeds on the one below it. Food chains show the flow of energy and nutrients through an ecosystem.

Food Web

A food web is a complex network of interconnected food chains. It shows the feeding relationships between different organisms in an ecosystem.

Ecological Pyramid

An ecological pyramid is a graphical representation of the trophic levels in an ecosystem. It shows the relative amounts of energy or biomass at each trophic level.

Balance Ecosystem

A balanced ecosystem is one in which the populations of different species are in balance with each other and with their environment. In a balanced ecosystem, the populations of predators and prey are stable, and the ecosystem is able to support a diverse range of species.

Effects of Human Activities on Environment

Human activities such as food production, shelter, housing, agriculture, industry, mining, transportation, economic and social security can have a significant impact on the environment. These activities can result in habitat destruction, pollution, and climate change.

Environmental Impact Assessment

Environmental Impact Assessment (EIA) is a process used to evaluate the potential environmental impacts of a proposed project or development. It is used to identify potential negative impacts and to develop measures to mitigate those impacts.

Sustainable Development

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It involves balancing economic, social, and environmental considerations in decision-making. Sustainable development is essential for the long-term health and well-being of both people and the environment.

Unit 2 - Natural Resources: Introduction, Classification

Natural resources are materials or substances that occur naturally within the environment and can be used for economic gain. They are classified into two main categories: renewable and non-renewable resources.

Renewable resources are those that can be replenished naturally in a relatively short period of time, such as sunlight, wind, and water. Non-renewable resources, on the other hand, are those that are finite and cannot be replenished, such as fossil fuels and minerals.

Water Resources

Water is one of the most important natural resources, as it is essential for life and many human activities. Water resources include surface water, groundwater, and rainwater.

Availability, sources and Quality Aspects

Water availability varies greatly depending on the location and climate. Some areas have abundant water resources, while others experience water scarcity. The main sources of water are rivers, lakes, and groundwater.

Water quality is also an important aspect of water resources. Water can be contaminated by pollutants, making it unsafe for consumption or use. Water treatment processes are used to improve water quality.

Water Borne and Water Induced Diseases

Water can transmit diseases when it is contaminated by pathogens. Waterborne diseases are caused by the ingestion of contaminated water, while water-induced diseases are caused by the lack of clean water for hygiene.

Fluoride and Arsenic Problems in Drinking Water

Fluoride and arsenic are two contaminants that can be found in drinking water. High levels of fluoride can cause dental and skeletal fluorosis, while high levels of arsenic can cause skin lesions and increase the risk of cancer.

Mineral Resources

Minerals are non-renewable natural resources that are extracted from the earth. They are used in a variety of industries, including construction, electronics, and energy production.

Material Cycles

Material cycles refer to the movement of elements and compounds through the environment. The carbon, nitrogen, and sulfur cycles are three important material cycles.

Carbon Cycle

The carbon cycle describes the movement of carbon between the atmosphere, oceans, and land. Carbon is taken up by plants during photosynthesis and released back into the atmosphere through respiration and decomposition.

Nitrogen Cycle

The nitrogen cycle describes the movement of nitrogen between the atmosphere, soil, and living organisms. Nitrogen is converted into various forms by bacteria and is taken up by plants to be used in the production of proteins.

Sulfur Cycle

The sulfur cycle describes the movement of sulfur between the atmosphere, soil, and living organisms. Sulfur is released into the atmosphere through volcanic eruptions and the burning of fossil fuels and is taken up by plants and incorporated into proteins.

Energy Resources

Energy resources are natural resources that can be used to produce energy. They are classified into two main categories: conventional and non-conventional sources of energy.

Conventional Sources of Energy

Conventional sources of energy are those that have been traditionally used to produce energy, such as fossil fuels and nuclear energy.

Non-Conventional Sources of Energy

Non-conventional sources of energy are those that are not traditionally used to produce energy, such as solar, wind, and geothermal energy.

Forest Resources

Forests are important natural resources that provide a variety of ecosystem services, including timber, carbon sequestration, and habitat for wildlife.

Availability, Depletion of Forests

Forests are found in many parts of the world, but their availability is decreasing due to deforestation. Deforestation is the conversion of forested land to non-forested land, usually for agricultural or urban development.

Environment impact of forest depletion on society

The depletion of forests has a significant impact on the environment and society. It can lead to soil erosion, loss of biodiversity, and changes in climate. It can also affect the livelihoods of people who depend on forests for their survival.

Unit 3 - Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid waste management.

Pollution is the introduction of harmful substances or products into the environment. It is a major problem in America and as well as the world. Pollution not only damages the environment, but it also damages the health of the people. There are several types of pollution, and they are categorized based on the part of the environment which they affect or result which the particular pollution causes. Each of these types has its own distinctive causes and consequences. Categorized study of pollution helps to understand the basics in more detail and produce protocols for the specific types.

Water Pollution

Water pollution is the contamination of water bodies, usually as a result of human activities. Water bodies include for example lakes, rivers, oceans, aquifers and groundwater. Water pollution results when contaminants are introduced into the natural environment. For example, releasing inadequately treated wastewater into natural water bodies can lead to degradation of aquatic ecosystems. In turn, this can lead to public health problems for people living downstream. They may use the same polluted river water for drinking or bathing or irrigation. Water pollution is the leading worldwide cause of death and disease, e.g. due to water-borne diseases.

Air Pollution

Air pollution is the release of pollutants into the air that are detrimental to human health and the planet as a whole. The Clean Air Act authorizes the U.S. Environmental Protection Agency (EPA) to protect public health by regulating the emissions of these harmful air pollutants. The NRDC has been a leading authority on this law since it was established in 1970.

Soil Pollution

Soil pollution is the presence of man-made chemicals or other alteration in the natural soil environment. This type of contamination typically arises from the rupture of underground storage tanks, application of pesticides, percolation of contaminated surface water to subsurface strata, oil and fuel dumping, leaching of wastes from landfills or direct discharge of industrial wastes to the soil. The most common chemicals involved are petroleum hydrocarbons, solvents, pesticides, lead and other heavy metals. This occurrence of this phenomenon is correlated with the degree of industrialization and intensities of chemical usage.

Noise Pollution

Noise pollution is the disturbing or excessive noise that may harm the activity or balance of human or animal life. The source of most outdoor noise worldwide is mainly caused by machines and transportation systems, motor vehicles, aircraft, and trains. Outdoor noise is summarized by the word environmental noise. Poor urban planning may give rise to noise pollution, side-by-side industrial and residential buildings can result in noise pollution in the residential areas.

Solid Waste Management

Solid waste management is the collection, transport, processing, recycling or disposal, and monitoring of waste materials. The term usually relates to materials produced by human activity, and the process is generally undertaken to reduce their effect on health, the environment or aesthetics. Waste management is a distinct practice from resource recovery which focuses on delaying the rate of consumption of natural resources. All waste materials, whether they are solid, liquid, gaseous or radioactive fall within the remit of waste management.

Unit 4 - Current Environmental Issues of Importance

Global Warming

- Global warming refers to the long-term rise in the average temperature of the Earth's climate system.
- It is primarily caused by human activities, such as the burning of fossil fuels and deforestation.
- These activities release large amounts of greenhouse gases into the atmosphere, which trap heat and cause the Earth's temperature to rise.

Greenhouse Effects

- The greenhouse effect is the process by which the Earth's atmosphere traps heat and warms the planet.
- This is a natural process that is essential for life on Earth, as it keeps the planet warm enough to support life.
- However, human activities have increased the levels of greenhouse gases in the atmosphere, which has enhanced the greenhouse effect and caused the Earth's temperature to rise.

Climate Change

- Climate change refers to the long-term changes in the Earth's climate, including changes in temperature, precipitation patterns, and wind patterns.
- It is caused by both natural factors and human activities, such as the burning of fossil fuels and deforestation.

- Climate change can have a wide range of impacts on the Earth's ecosystems, including more frequent and intense heatwaves, droughts, and storms.

Acid Rain

- Acid rain is a type of precipitation that is acidic in nature.
- It is caused by the release of sulfur dioxide and nitrogen oxides into the atmosphere, which react with water, oxygen, and other substances to form sulfuric acid and nitric acid.
- Acid rain can have harmful effects on plants, aquatic animals, and infrastructure.

Ozone Layer Formation and Depletion

- The ozone layer is a layer of ozone molecules in the Earth's stratosphere that protects the planet from harmful ultraviolet radiation from the sun.
- Ozone is formed when oxygen molecules are broken apart by ultraviolet radiation and the resulting atoms recombine with other oxygen molecules to form ozone.
- However, human activities, such as the release of chlorofluorocarbons (CFCs) into the atmosphere, have depleted the ozone layer, allowing more harmful ultraviolet radiation to reach the Earth's surface.

Population Growth and Automobile Pollution

- The rapid growth of the human population has put a strain on the Earth's resources and ecosystems.
- One of the impacts of population growth is the increase in the number of automobiles on the road, which has led to an increase in air pollution.
- Automobiles release pollutants, such as carbon monoxide, nitrogen oxides, and particulate matter, into the air, which can have harmful effects on human health and the environment.

Burning of Paddy Straw

- Paddy straw is the residue left after the harvesting of rice crops.
- In some regions, farmers burn paddy straw to quickly clear their fields for the next crop.
- However, the burning of paddy straw releases large amounts of pollutants into the air, including particulate matter, carbon monoxide, and nitrogen oxides, which can have harmful effects on human health and the environment.

Unit 5 - Environmental Protection

Environmental Protection Act 1986

- The Environmental Protection Act of 1986 is an act of the Parliament of India.
- The purpose of the act is to protect and improve the environment and prevent hazards to human beings, other living creatures, plants, and property.
- The act provides for the establishment of authorities at the national and state levels to regulate environmental pollution.
- The act also provides for penalties for non-compliance with its provisions.

Initiatives by Non-Governmental Organizations (NGOs)

- NGOs play a crucial role in environmental protection and conservation.
- They work to raise awareness about environmental issues and promote sustainable development.
- NGOs also engage in advocacy and lobbying to influence government policies and actions.
- Some well-known NGOs working in the field of environmental protection include Greenpeace, World Wildlife Fund, and Friends of the Earth.

Human Population and the Environment

- Population growth has a significant impact on the environment.
- As the population grows, the demand for resources such as food, water, and energy increases.
- This can lead to overexploitation of natural resources, deforestation, and pollution.
- Environmental education is essential to raise awareness about the impact of human activities on the environment.
- Women's education is particularly important as it can help to reduce population growth and promote sustainable development.